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First Examiner J. BERRIA	Paper Code E3.17
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(Q1a) a

Home system : Network to which mobile user's permanent phone number belongs

Home MSC / Location Reg. HLR :
- MSC : point of contact to obtain address of mobile user
- HLR : database in home system containing permanent information of mobile user

Visited system : Network other than home system where mobile user is currently residing

Visited MSC / Visitor Location Reg. VLR :
- MSC : responsible for setting up calls to/from mobile nodes
- VLR : temporary database entry in visited system for each visiting mobile user

Mobile station Roaming Nr. (MSRN) : routable address for telephone call segment between home MSC and visited MSC

5

(Q1b) a

Home system ↔ Home network

Home MSC / HLR ↔ Home agent

Visited system ↔ Visited Network

Visited MSC / Visitor Location Reg VLR ↔ Foreign Agent

Mobile station Roaming Nr (MSRN) ↔ Care of address

3

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Q1 b)

Broadcast network: Data transmitted by one station received by many.
Stations share common transmission media
Media Access control (MAC) mechanism needed
point to point link: one sender, one receiver, one link.
No media access control (MAC) mechanism

4

channel partitioning: Allocate capacity of media to a node exclusively: Examples: time division multiplexing and frequency division multiplexing.
Round Robin: Node take turns to access media. Example: Token ring (IEEE 802.5)
Random Access: There are contention mechanisms. The MAC protocol must specify How to detect collisions, & How to recover & from collision. Examples: CSMA, CSMA-CD, CSMA-CA

4

Flow control
- Assures that a sender station does not overwhelm a receiver station

Examples:
- Stop and wait ARQ
- Selective repeat ARQ
- Go Back N ARQ

Introduction to Window based

4

Q2 a)

$T = x(1) + x(2)^2$

$R = x(1) + x(2)$

$\frac{\partial T}{\partial x(1)} = \frac{\partial T}{\partial x(2)} \qquad 1 = 2x(2)$ 4

$\frac{\partial T}{\partial x(1)} = 1$. This means that link 1 has the same cost regardless of how much flow is assigned to it.

$\frac{\partial T}{\partial x(2)} = 2x(2)$. link 2 has a cost that is proportional to the flow 4

$1 = 2x(2) \Rightarrow x^*(2) = \frac{1}{2}$

$x^*(1) = R - \frac{1}{2}$

4

Q2 b)

Reliable data transfer sequence : Stop and Wait

L = length of segment in bits
R = transmission rate of channel in bps
RTT = Round Trip time in s

2

$$U = \frac{L/R}{RTT + L/R}$$

2

$$U = \frac{0.009}{20 + 0.009} = \frac{0.009}{20.009} = 0.00045$$

2

Sliding windows protocols :
sender is allowed to send multiple packets without acknowledgement
For example if we allow for three packet without acknowledgement the Utilization is given by :

$$U = \frac{3 \times 0.009}{20.009} = 0.0135$$

2

Q3 a)

Network assisted congestion control:
- Routers provide feedback to end system. For example single bit indicating congestion or explicit rate for sender to send at.

End to end congestion control:
- No explicit feedback from e.g. network routers
- congestion inferred from end-system observed loss or delay

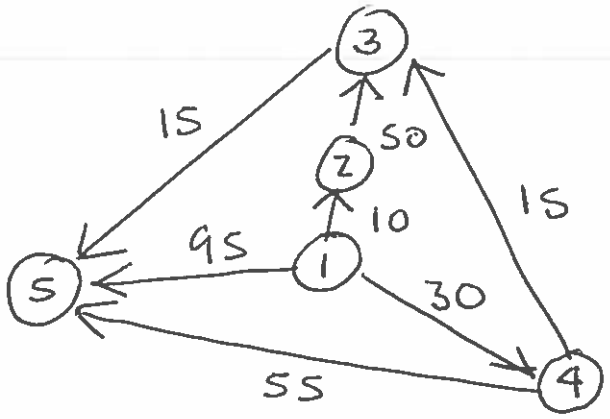
This is the approach taken by TCP protocol. 4

TCP congestion control

Additive increase and multiplicative decrease
- sender increases transmission rate (window size), probing for usable bandwidth until loss occurs
- TCP slow start: when connection begins, increase rate exponentially until first loss event
- If a loss event is detected, the congestion window is decreased to 1 MSS (maximum segment size); or the window is reduce to half window size previous loss. 3

Fairness goal: If K TCP session share same bottleneck link of bandwidth R each should have given an average rate of R/K. 3

Q3 n)



$i=0 \quad S=\{1\}$, $D(2)=10$, $D(3)=\infty$, $D(4)=30$, $D(5)=95$

$i=1 \quad S=\{1,2\}$

$D(3) = 60 = \min(\infty, D(2)+c(2,3))$

$D(4) = 30 = \min(30, D(2)+c(2,4))$

$D(5) = 95 = \min(95, D(2)+c(2,5))$ 5

$i=2 \quad S=\{1,2,4\}$

$D(3) = 45 = \min(60, D(4)+c(4,3))$

$D(5) = 85 = \min(95, D(4)+c(4,5))$

$i=3 \quad S=\{1,2,4,3\}$

$D(5) = 60 = \min\{85, D(3)+c(3,5)\}$

$i=4 \quad S=\{1,2,4,3,5\}$ 5

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Q4 a) i) MPLS separate networks in their own right:
MPLS - Integrates layer 2 switching with layer 3 routing. It operates multiple layer 2 technologies
MPLS does not define another QoS architecture
- MPLS uses DiffServ architecture
- MPLS scalability: aggregation of traffic at edge router. At core router processing of aggregate of traffic only. 3

ii)

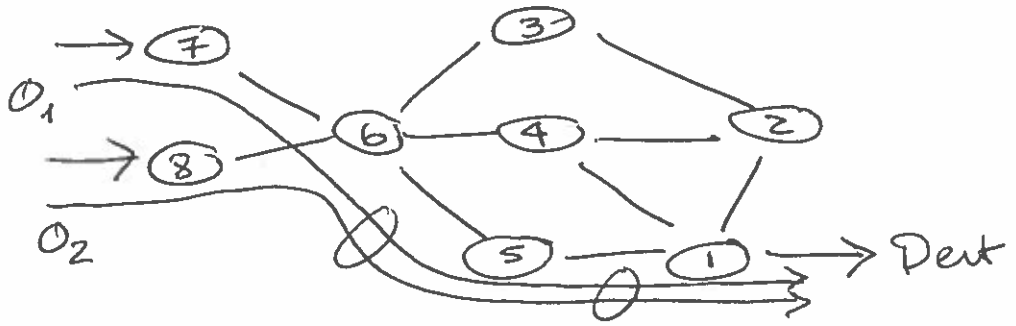
Routing protocol } control plane
Routing table
↓
Forwarding Tables } Forwarding plane
Switching Fabric

Key feature in MPLS is the separate of control plane and forwarding data plane
control plane: set up route through data plane
data plane: carries information 3

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Q4 a iii)

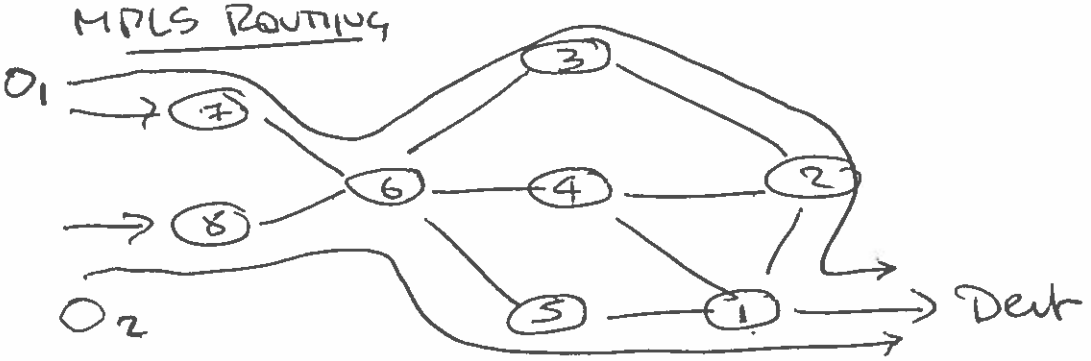
IP ROUTING



IP Routing : path to destination is determined by destination address alone

2

MPLS ROUTING

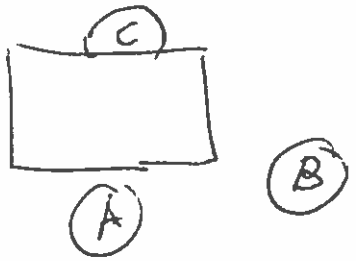


MPLS Routing : path to destination can be based on source and destination address

For example : pre-store back up route in case of link failure

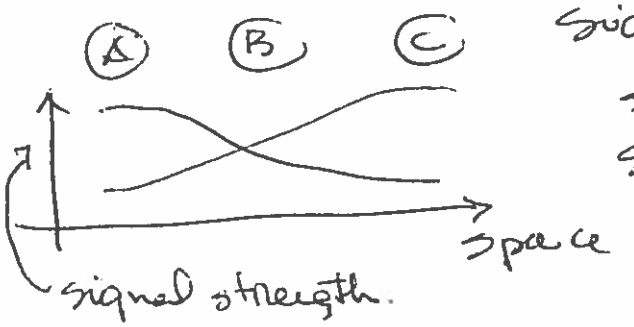
2

Q4 b) i)



Hidden Terminal Problem:
stations A & B hear each other
stations B & C hear each other
But stations A & C cannot hear each other.

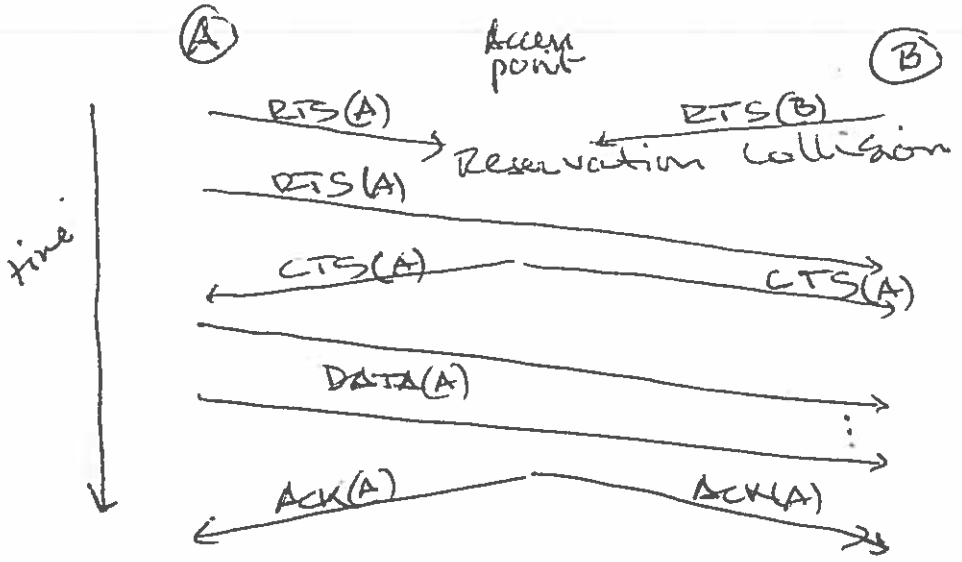
Therefore, station A and C are unaware of their interference at B. 2



Signal attenuation:
stations A & B hear each other
Stations B & C hear each other

But station A & C cannot hear each other interfering at station B. 2

Q4 (a) (ii)



3

The idea is to allow the sender to reserve the channel

- Senders first transmits a small request-to-send (RTS) packet to the Access point (Base station) using Carrier Sense multiple access technique
- If the request was successful the Access point (Base station) broadcast a clear-to-send (CTS) packet. This CTS packet is heard by all nodes
- Sender transmit data frame.
- All other stations defer transmissions

3